Journal of Biomedical Informatics 129 (2022) 104069

Contents lists available at ScienceDirect

Journal of Biomedical Informatics

journal homepage: www.elsevier.com/locate/yjbin

Original Research

TAHDNet: Time-aware hierarchical dependency network for medication recommendation

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ARTICLE INFO

Keywords:
Time-aware
Medication recommendation
Attention mechanism

ABSTRACT

Medication recommendation is a hot topic in the research of applying neural networks to the healthcare area. Although extensive progressions have been made, current researches still face the following challenges: (i). Existing methods are poor at efficiently capturing and leveraging local and global dependency information from patient visit records. (ii). Current time-aware models based on irregularly interval medical records tend to ignore periodic variability in patient conditions, which limits the representational learning capability of these models. Therefore, we propose a Dynamic Time-aware Hierarchical Dependency Network (TAHDNet) for the medication recommendation task to address these challenges. Firstly, we use a Transformer-based model to learn the global information of the whole patient record through a self-supervised pre-training process. Secondly, a 1D-CNN model is used to learn the local dependencies on visitation level. Thirdly, we propose a dynamic time-aware module with a fused temporal decay function to assign different weights among different time intervals dynamically through a key-value attention mechanism. Experimental results on real-world datasets demonstrate the effectiveness of the model proposed in this paper.

1. Introduction

In recent years, with a large amount of electronic health record (EHR) data widely available, neural networks have demonstrated great success in prediction tasks of the healthcare area. Recommending medication accurately and effectively is crucial to assist doctors in making decisions. Currently, plenty of deep learning models were proposed to apply to the problem [1–6]. These methods learn representations of medical entities (e.g., patients, diagnosis, medications) from patient EHR data and then use the generated new representations to achieve the medical prediction tasks. To learn representation accurately, various longitudinal recommendation methods treat the recommendation task as a sequential prediction task [7–11]. Other methods are instance-based, such as LEAP [6], focus on the current health conditions of each patient. Besides, some methods leverage the temporal dependencies among clinical events, see [8,2,10,12].

However, the medication recommendation methods based on neural network have two limitations: (1) Current methods mainly focus on capturing global information (which refers to the overall representation of the patient records.) or local information (which refers to the local representation of patient records learned through a sliding window.) [13] of patient's records, without considering the combination of both information. This is because in the process of making medication recommendations, doctors sometimes need to consider not only the patient's recent medical visits, but also the patient's overall medical visits to make appropriate medication recommendations. (2) Existing approaches rely on the assumption that medical visits with the same time interval have the same impact on disease progression, using a homogeneous decay to attenuate the effects of previous time steps on the current time step, which is contrary to the actual situation of disease progression in sometimes [14,15]. As shown in Fig. 1, a pneumonia patient's medical procedure sequence is known to predict his medication in the next time. The time interval between visit sequences varies from 15 months to 3 days. The medications of the patient's visit 1 were penicillin and cephalosporins. The medications for visit 2 and visit 3 were both penicillin and β-lactams. In Fig. 1, the global dependency

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https://doi.org/10.1016/j.jbi.2022.104069
Received 22 September 2021; Received in revised form 15 March 2022; Accepted 31 March 2022
Available online 4 April 2022
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information refers to learning the overall representation of Visit1, Visit2 and Visit 3 (marked by the black dashed line). The local dependency information refers to learning the local representation of the visit record by setting the sliding window. For example, if we set the sliding window to 2, we will get the local representation by learning Visit 1 and Visit 2, and Visit 2 and Visit 3 (marked by the red dashed box). It can be seen from Fig. 1 that doctor believe that the medication situation of 3 days ago may be more important for medication recommendation in the future. In addition, as the patient's condition changes, the medication situation also changes.

To alleviate the above mentioned limitations, We propose a novel Time-Aware Hierarchical Dependency Network (TAHDNet), an end-to-end deep learning model for medication recommendation. The main contributions of TAHDNet are summarized as below:

- We propose a hierarchical dependence learning method, the main idea of which is to capture the dependence information between patients and medications from both global[1] and local levels.[2] Specifically, on the one hand, we use an adapted Transformer-based learning module to learn the global representation of the whole patient record through a self-supervised pre-training process. On the other hand, we learn the local information of the patient record by using a sliding window in 1D-CNN. By learning aforementioned the global and local hierarchical dependencies, we can learn accurate information of patients.
- We propose a module with a fused temporal decay function and assign weights to different time steps dynamically through a key-value attention mechanism. We notice that some diseases develop in a cyclical manner [14,15], which means handle time intervals only in a monotonic decay function will lead to suboptimal results. For this consideration, we propose a fused temporal decay function consists of periodic decay and monotonic decay for a dynamic time-aware attention module to model the disease progression. In this way, we can evaluate disease progression more realistic.
- Considering the impact of Drug Drug interactions (DDI) on the recommended performance of the model, we added the DDI loss on the basis of the TAHDNet model to jointly determine the recommended results.

The rest of this paper is organized as follows. In Section 2, we briefly review related work on medication recommendation, hierarchical dependency, and time-aware attention models. In Section 3, we present the definitions related to the task of this paper. In Section 4, we describe the main modules of the proposed model, TAHDNet. We then describe the experimental implementation details, experimental results and ablation experiments in Section 5. Finally, we conclude the work of this paper in Section 6.

2. Related work

In this section, we present related work on medication recommendation, hierarchical dependence, and time-aware attention.

2.1. Medication recommendation

Most of the traditional methods use tensor, bayesian or regression models to implement medication recommendation. For example, Wang et al. [16] proposed a drug combination prediction framework based on a polynomial regression model, in which the function learning problem is transformed into a tensor reconstruction problem to improve the time efficiency of the model. Chen et al. [17] proposed a DrugCom architecture based on tensor decomposition, which integrates multiple heterogeneous data sources of drugs and diseases to calculate drug combinations for different diseases, and uses an alternating direction multiplier algorithm to effectively solve the optimization problem. Muhammad et al. [18] proposed a sparse linear regression model based on Bayesian multi-view multi-task to realize drug combination prediction. This method predicts a group of highly correlated drug combinations by identifying feature combinations from multiple input data sources and considering the correlation effects between drugs. He et al. [19] converted personalized drug recommendation into a ranking problem, and proposed a method called Kernelized Ranking Learning (KRL), which directly optimizes the clinically relevant scenarios of personalized drug recommendation and predicts top-ranked drugs. Although these methods have achieved good results, they have not fully explored the relationship between drugs and the regularity of patients' medication. Recently, with the continuous development of computer technology, various of deep learning models have been proposed for medication recommendation [20,21]. Medication recommendation can be categorized into instance-based and longitudinal recommendation methods [3]. Instance-based methods focus on current health conditions. Among them, Leap [6] uses a recurrent decoder to model label dependency and uses content-based attention to model the underlying label-to-instance mapping to predict a combination of medicines given a patient's diagnosis. Longitudinal-based methods leverage the temporal dependencies among clinical events, see [8,2,10].

2.2. Hierarchical dependency

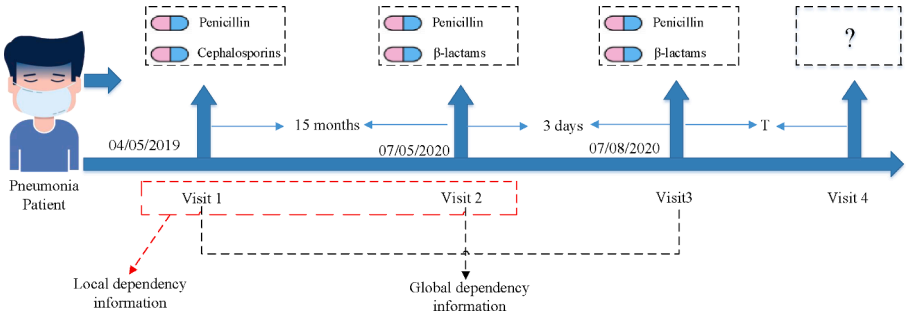


Fig. 1. There is a medication treatment process for patients with pneumonia.

Hierarchical dependency refers to capturing the interaction information of medical visit records from different perspectives to enhance the representational power of the data, so as to better complete medical tasks (such as mortality risk prediction, ICU transfer prediction, disease prediction, medication recommendation, etc.). The hierarchical dependency mentioned in this paper refers to the medical information

[1] Global dependency information refers to the overall information representation of the patient record.

[2] Local dependency information refers to learning local information representation about the patient's record through a sliding window.

mining from the local and global levels. With the introduction of attention mechanism [22] to neural networks, various of attention-based models are proposed to learn to assign different weights for each visit and generate a weighted representation for each patient [23–27]. Zhou et al. [13] proposed a Global attention mechanism (GATT) to measure the local contribution of each input activity towards the combined sequence representation to achieve the recommendation task. RETAIN [8] trains two RNN in a reverse time order to efficiently generate the appropriate attention variables. RETAIN mimics physician practice by attending the EHR data in reverse time order so that recent clinical visits are likely to receive higher attention weight. INPREM [25] is designed as a linear model for interpretability while encoding non-linear relationships into the learning weights for modeling the dependencies between and within each visit. It enables the model to obtain the input variables' contribution matrix, which is the evidence of the prediction result(s). Dipole [28] is an end-to-end, simple, and robust model for predicting patients' future health information. Dipole introduces three attention mechanisms to measure the relationships of different visits for the prediction. Besides traditional neural networks, AMANet [26] integrates both attention and memory to solve asynchronous multi-view learning problems in general. SAnD [24] employs a masked, self-attention mechanism and uses positional encoding and dense interpolation strategies for incorporating temporal order.

2.3. Time-aware attention models

Most neural network methods focus on learning to model the sequential characteristics of EHR data. However, EHR data are not only sequential but also temporal because each visit is associated with time information, which is highly related to prediction tasks [29]. Besides, the time intervals between neighbor visits are irregular, how to utilize the irregular time information becomes a crucial problem. T-LSTM [2] is proposed to handle irregular time intervals in longitudinal patient records. T-LSTM learns a subspace decomposition of the cell memory, enabling time decay to discount the memory content according to the elapsed time. RetainEX [30] is proposed based on the same assumption with T-LSTM that time decay discounts memory. RetainEX is built upon Retain by considering information decay, which also uses traditional attention mechanisms to learn weights for visits and diagnosis ontology. HiTANet [29] proposes a new time-aware key-query attention mechanism to identify the critical time steps among patients' historical visits in their EHR data. It uses a patient's overall representation as a query vector and the time embedding of each visit as the key vector.

In summary, current approaches tend to ignore local dependency learning and the cyclical nature of certain disease progressions. In contrast, we propose a Dynamic Time-aware Hierarchical Dependence Network(TAHDNet), which performs hierarchical dependence learning of global and local dependencies and uses a fused temporal decay function to handle time intervals, resulting in more realistic learning of disease progression.

3. Task definition

Definition 1. (*Longitudinal EHR Data*) In longitudinal patient records, we can represent each patient's records as a multivariate sequence: $X^{(n)} = \left[X_1^{(n)}, X_2^{(n)}, \cdots, X_{T^{(n)}}^{(n)}\right]$, in which $n \in 1, 2, \cdots, N, N$ is the total number of patients and $n$is the n-th patient, $T^{(n)}$ is the number of visits of the n-th patient. For simplicity, we will describe the algorithms for a single patient and omit the superscript whenever it is unambiguous. Thus, $X_t = C_t^d \bigcup C_t^m$ is the concatenation of medical codes of the t-th visit of a single patient, where $C_t^d \subset C^d$ is diagnoses codes (we use ICD-9.) and $C_t^m \subset C^m$ is medication codes (we use ATC). To avoid duplicating, we use a multi-hot vector $C_t^* \in \{0,1\}^{|C^*|}$ to represent the medical codes we mentioned before, where C^* indicates each medical code set and $|C^*|$ indicates the size of each code set.

Definition 2. (*Time Interval*) For each patient, we calculate the time intervals (in days) of two adjacent visits. The intervals embedding can be represented as $\delta^{(n)} = [\delta_1^{(n)}, \delta_2^{(n)}, \cdots, \delta_{T^{(n)}}^{(n)}]$, N is the total number of patients and $n$is the n-th patient, $T^{(n)}$ is the number of visits of the n-th patient. We also omit the superscript $n$the same way we describing the record data. To calculate each interval, let $\delta_n = W_\delta(t_n - t_{n-1}) + b_\delta$, where δ_n represents the embedding time interval between the n-th visit and the previous visit. Since the first visit has no time interval, we set $\delta_1 = 0$.

Problem Definition (Medication Recommendation). For each patient, given patient's visit history before time t, $X_{1:\,t-1} = [X_1, X_2, \cdots, X_{t-1}]$, and diagnosis code C_t^d of each visit at time t. Our goal is to recommend the possible medications for each patient at time t. We use a multi-label output $\hat{y}_t \in \{0,1\}^{|C^m|}$, $|C^m|$ indicates the size of medication codes.

4. Method

The overall framework of TAHDNet is illustrated in Fig. 2. It contains three main blocks: 1D-CNN block, Transformer block, and Time-aware block. The model learned hierarchical dependencies on longitudinal EHR data by 1D-CNN block(for local dependency) and Transformer block(for global dependency), and dynamic time-aware attention by Time-aware block. The outputs of these three blocks are concatenated to generate a new representation for each patient, and then the new representation is inputted into the prediction layer for the further medication recommendation task. In addition, we introduced Drug Drug Interactions (DDI) loss to co-determine the final recommendation. The detailed introduction of the model is as follows:

4.1. Transformer block

Inspired by the recent achievements of the Transformer model on natural language processing and sequence transduction, some recent works introduce Transformer structure into the healthcare domain [31,32] to learn the interactions of medical records. Based on that, we adapt the Transformer structure and use a pre-trained Transformer-based module to model the global dependency information on the whole patient records. Our reasons for choosing a Transformer-based structure are twofold. On the one hand, each visit's diagnosis codes and medication codes are unordered. Therefore order-based models such as RNNs [33,7,34] may not be suitable for this situation. On the other hand, RNNs calculate interactions in a black box, which impairs the model's interpretability. By using a multi-head self-attention mechanism, Transformer can measure the interactions of two medical records intuitively (see Fig. 2).

In this part, we follow the work of G-BERT [32] to adapt the Transformer structure in the following aspects:

Pre-Training

We adopt the pre-training method that G-BERT [32] uses. For self-prediction task, we pre-train the model to minimize the binary cross-entropy loss $\mathcal{L}_{se}$ as follows:

$$\mathcal{L}_{se}(h_t,\ C_t^*) = -logp(C_t^*|h_t) = -\sum_{c \in C_t^*} \log p(c_t^*|h_t) + \sum_{c \in C^* \setminus C_t^*} \log p(c_t^*|h_t) \tag{1}$$

In practice, we use a specific symbol [MASK] to randomly replace 15% of the original medical code $c^* \in C_t^*$. So the model should have the ability to predict the masked code based on others. Likewise, for the dual-prediction task, we pre-train the model by optimizing the binary cross-entropy loss $\mathcal{L}_{du}$as follows:

$$\mathcal{L}_{du} = -\log p(C_t^d|h_t^d) - \log p(C_t^m|h_t^m) \tag{2}$$

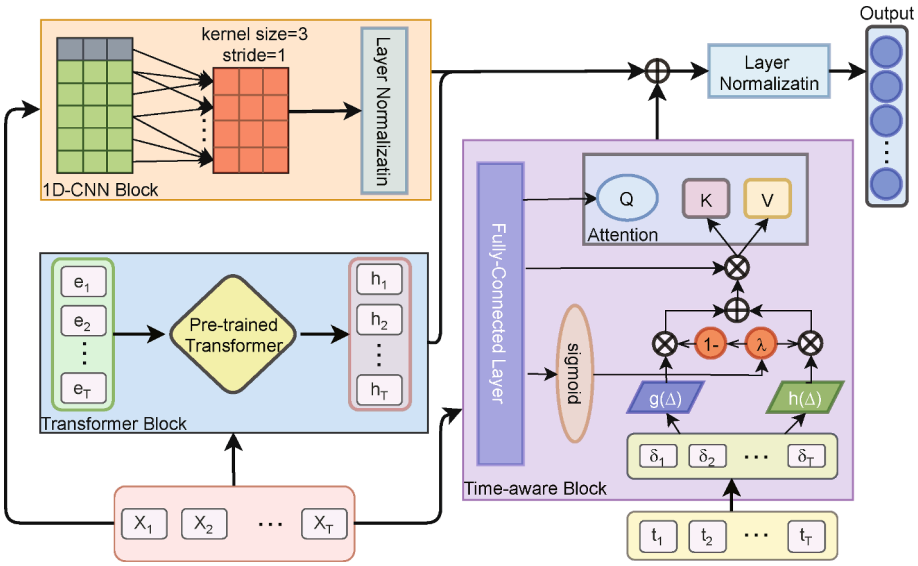


Fig. 2. The architecture of TAHDNet.

Our final pre-training optimization objective is the combination of the losses above:

$$\mathcal{L}_{pre} = \frac{1}{N}\sum_{n=1}^{N}\left(\mathcal{L}_{se}\left(h_t^{(n)},\ C_t^{d}\right) + \mathcal{L}_{se}\left(h_t^{(n)},\ C_t^{m}\right) + \mathcal{L}_{du}^{(n)}\right) \tag{3}$$

Visit Embedding

Firstly, we embed the visit vector to the latent visit space using a linear function as follows:

$$e_t = W_e X_t + b_e \tag{4}$$

where $e_t \in \mathbb{R}^{\widetilde{h}}$, $\widetilde{h}$ is hidden size, $W_e \in \mathbb{R}^{\widetilde{h}\times N}$ is the weight matrix, and $b_e \in \mathbb{R}^{\widetilde{h}}$ is the bias vector. By this way, the input data of each patient can be represented by $E = [e_1, e_2, \cdots, e_T]$. Then we use a pre-trained Transformer network to learn the interactions among medical ontologies as follows:

$$h_T = Transformer(e_1, e_2, \cdots, e_T) \tag{5}$$

where $h_T \in \mathbb{R}^{\widetilde{h}}$ is the latent space representation with global dependencies.

It is worth noting that there are two differences between our Transformer block and G-BERT [32]. Firstly, G-BERT uses ontology (e.g., patients, diagnosis, medications) embedding output as input, but we use a linear embedding result of visit as input to learning better representation. Secondly, G-BERT uses the Transformer output for final prediction, but we use it only as the global dependency representation. We then fuse it and the other two representations of 1D-CNN block and Time-aware block to generate the final representation for the medication prediction task.

4.2. 1D-CNN block

Though most medical prediction models [8,2,9,10] are built upon recurrent neural networks(RNN, LSTM, GRU, etc.) to calculate the interactions among different visits, we choose 1D-CNN to model the local dependency information. We make this choice rely on the situation that doctors tend to focus more on the more recent visits than the long-ago ones when a patient goes to the hospital. In this part, we introduce how to model interactions between the current visit and its recent visits.

1D-CNN block takes the multivariate sequence of visit $[X_1, X_2, \cdots, X_T] \in \mathbb{R}^{T\times|C^*|}$ as the inputs, and we use a 1D-CNN network to learn the dependencies between neighbor visits. In this way, we model the local dependency information. The procedure embedding is as follows:

$$h_C^{'} = CNN_{1d}(X_1, X_2, \cdots, X_T) \tag{6}$$

where $h_C^{'} \in \mathbb{R}^{\widetilde{h}\times|C^*|}$ is the hidden layer output of 1D-CNN network, meanwhile $\widetilde{h}$ is hidden size, a hyperparameter of TAHDNet.

To avoid Internal Covariate Shift, we introduce layer normalization into 1D-CNN block, which is introduced next:

$$h_C = LayerNorm\left(h_C^{'}\right) = \alpha \odot \frac{x-\mu}{\sqrt{\sigma^2+\epsilon}} + \beta \tag{7}$$

where μ is the mean value of a layer, σ^2 is the variance of a layer, α and β represent the parameter vectors for scaling and translation, respectively.

4.3. Time-aware block

Traditional time-aware approaches [2,29,30] process time intervals under the assumption that as the time intervals increase, the effect of past visits to the current visit will decay. However, this assumption does not correspond to the actual situation. The progression of some diseases is cyclical, which leads to cyclical diagnosis and medications. Thus, monotonic decay may not suites this reality accurately. We propose a Time-aware block to model the irregular time intervals to generate a better representation for the prediction task.

In Time-aware block, we first propose a fused decay function to consider both periodic decay and monotonic decay. We use an interval gate $\lambda$to fuse the two decay functions. The specific formulas are as follows:

$$q_t = W_f X_t + b_f \tag{8}$$

$$\lambda = Sigmoid(q_t) \tag{9}$$

where $W_f \in \mathbb{R}^{\widetilde{h}\times|C^*|}$ are parameters same as Eq. (4), $q_t \in \mathbb{R}^{\widetilde{h}}$ is the time-aware query vector. $\lambda \in \mathbb{R}^{\widetilde{h}}$ is the interval gate to fuse the two time decay functions as bellow:

$$\delta_t = ReLU(W_t t_t + b_t) \tag{10}$$

$$g(\Delta) = \sin\delta_t \tag{11}$$

$$h(\Delta) = tanh\left(\frac{1}{\log(e+\delta_t)}\right) \tag{12}$$

$$k_t = v_t = q_t \otimes (\lambda \times h(\Delta) + (1-\lambda) \times g(\Delta)) \tag{13}$$

where $\delta_t \in \mathbb{R}^{\widetilde{h}}$ is the time interval embedding vector through a nonlinear activation function ReLU to only keep the positive values. Here we use two different decay functions to decay the time intervals, a periodic decay function $g(\Delta)$, and a monotonic decay function $h(\Delta)$. We use λ to fuse the two decay functions aforementioned for the overall discount function. In this way, we obtain the time-aware key vector and value vector.

Following the self-attention mechanism proposed in Transformer [22], we calculate the attention weights and generate the latent space representation of time intervals as follows:

$$\phi_t = Attention(Q,K,V) = \frac{QK^T}{\sqrt{d_k}}V \tag{14}$$

$$h_I = LayerNorm(\phi_1, \phi_2, \cdots, \phi_T) \tag{15}$$

where Q is the matrix composed of $[q_1, q_2, \cdots, q_T]$, K is the matrix composed of $[k_1, k_2, \cdots, k_T]$, and V is the matrix composed of $[v_1, v_2, \cdots, v_T]$.

4.4. Prediction

Based on the latent space representation we obtained before, we finally concatenate them to obtain the representation of a patient record as follows:

$$h^{'} = Concat(h_T, h_C, h_I) \tag{16}$$

where $h^{'} \in \mathbb{R}^{5\times\widetilde{h}}$. Finally, we use an MLP base prediction layer to predict the recommendation medication codes as follows:

$$\widehat{y}_t = ReLU\left(W_p h^{'} + b_p\right) \tag{17}$$

where $W_p \in \mathbb{R}^{|C^m|\times 5\widetilde{h}}, b_p \in \mathbb{R}^{|C^m|}$ are the learnable parameters. Given the true labels $y_t \in \{0,1\}^{|C^m|}$ at each time stamp t we use cross-entry loss as the objective function:

$$\mathcal{L} = -\frac{1}{T-1}\sum_{t=2}^{T}\left(\widehat{y}_t^T \log y_t + \left(1-\widehat{y}_t^T\right)\log(1-y_t)\right) \tag{18}$$

In addition, considering the impact of Drug Drug Interactions (DDI) [3] on the recommended performance of the model, we added the DDI loss to jointly determine the recommended results. The DDI loss ($\mathcal{L}_{loss}$) is calculated as follows:

$$\mathcal{L}_{loss} = \sum_{t}^{T}\sum_{i,j}\left(A_l \odot \left(\widehat{y}_t^T y_t\right)\right)[i,j] \tag{19}$$

Among them, every element in $\widehat{y}_t^T y_t \in \mathbb{R}^{N\times N}$ gives the pair-wise probability of predicted result. $\odot$ is the element-wise product. A_l represents an adjacency matrix composed of 0 and 1. If ij combined to induce a DDI, then $A_l[i,j] = 1$.

Therefore, the loss in this paper consists of two parts: the recommended task ($\mathcal{L}$) loss and the DDI loss ($\mathcal{L}_{loss}$). The calculation is as follows:

$$loss_{finall} = \mathcal{L} + \pi\mathcal{L}_{loss} \tag{20}$$

where π denotes the learnable hyperparameter.

5. Experiment

This section introduces the experimental setting, including dataset, baselines, and implementation details. Then we conclude the experimental results and discuss the effectiveness of TAHDNet.

5.1. Dataset

We used EHR data from MIMIC-III [35] and conducted all our experiments on a cohort where patients have more than one visit. As for patients who have only one visit, we utilize their data as pre-training data to avoid selection bias and enrich the pre-training dataset. We handle data in the same way as G-BERT [32] but reserve the time of each visit. The statistics of the datasets are summarized in Table 1.

5.2. Experimental setting

Firstly, we use the data of single visit patients to pre-train the Transformer block for parameters initializing. Then we randomly divide the dataset of multiple visit patients into training, validation, and testing set at a rate of 0.6: 0.2: 0.2, and experiment with the model according to the cross-validation method. The hyperparameter setting of TAHDNet is the same as G-BERT [32]: the dimension of the hidden layer as 300, dimension of position-wise feed-forward networks as 300, 2 hidden layers with 4 attention heads for each layer. We pre-train the model with 100 epochs to stabilize the initial parameters. We train the model with 100 epochs to evaluate the performance of TAHDNet. All experiments were performed in the environment as in Table 2.

In addition, We choose Jaccard Similarity Score(Jaccard), Average F1 Score(F1), and Precision Recall AUC(PR-AUC) as the metrics to measure the prediction accuracy. Jaccard is defined as the size of the intersection divided by the size of the union of ground truth set $Y_t^{(k)}$ and predicted set $\widetilde{Y}_t^{(k)}$. Precision can reflect the false alarm rate, while Recall can reflect the rate of the missing report, and F1 is the average score of Precision and Recall, PR-AUC is the area under the Precision-Recall curve.

5.3. Baselines

We compared TAHDNet with the following baselines. Since we handle data in the same way as G-BERT [3], we quote the baselines and experimental results of G-BERT.

Logistic Regression (LR) is logistic regression with L1/L2 regularization. Here we use a summary of the multi-hot vector of each visit to represent sequential multiple medical codes. We use the binary relevance technique [29] to handle the multi-label output.

LEAP [6] is an instance-based medication combination recommendation method that formalizes the task in a multi-instance and multi-label learning framework. A recurrent decoder is used to model label dependencies, and content-based attention is used to learn the disease-

Table 1
Statistics of the Data (dx: diagnosis code, rx: medication code).

Stats	Single-Visit	Multi-Visit
# of patients	30744	6351
avg # of patients	1	2.36
avg # of dx	10.82	10.51
avg # of rx	13.78	8.82
# of unique dx	1997	1958
# of unique rx	323	145
avg # of visit	1	2.36

Table 2
Detailed configuration of the server.

Operating System	Ubuntu 20.04.2 LTS
RAM	256 GB DDR4 @ 3200 MHz
CPU	AMD Ryzen Threadripper 3970X @ 3.70 GHz
GPU	NVIDIA GeForce RTX 3090 × 2
SSD	2.0 TB (NVM Express, PCIe 4.0)
Software	NVIDIA CUDA 11.2, PyTorch 1.8.1

drug mapping.

RETAIN [8] trains two RNNs in a reverse time order to efficiently generate the appropriate attention variables. It improves the prediction accuracy through a sophisticated attention generation process while keeping the representation learning part simple for interpretation, making the entire algorithm accurate and interpretable.

GRAM [36] supplements electronic health records (EHR) with hierarchical information inherent to medical ontologies (e.g., patients, diagnosis, medications). Based on the data volume and the ontology structure, GRAM represents a medical concept as a combination of its ancestors in the ontology via an attention mechanism.

GAMENet [3]integrates the drug-drug interactions knowledge graph by a memory module implemented as a graph convolutional network, using longitudinal patient records as the query. For a fair comparison, G-BERT uses a variant of GAMENet without DDI knowledge and procedure codes as input renamed as GAMENet- [32].

G-BERT [32] is the method that combines the power of Graph Neural Networks (GNNs) and BERT (Bidirectional Encoder Representations from Transformers) for medical code representation and medication recommendation. G-BERT also evaluated 3 variants for model ablation.

G −BERT$_{G^-,P^-}$ [32]: Medical embedding without input medical ontology (e.g., patients, diagnosis, medications) information and initializing the model's parameters without pre-training.

G −BERT$_{G^-}$ [32]: Medical embedding without input medical ontology information and initializing the model's parameters with pre-training.

G −BERT$_{P^-}$ [32]: Medical ontology embedding with medical ontology information input and initialize the model's parameters without pre-training.

5.4. Results

We evaluate the performance of TAHDNet on the medication recommendation task by comparing it with other classic methods. As we can observe from Table 3, TAHDNet achieves the top scores on all the metrics we use.

Among all the baselines, LR is worse than other models in terms of most metrics. The reason is that LR lacks the ability to model sequentially. Attention-based models, GRAM and RETAIN outperformed classic models. However, attention-based methods usually aggregate all visits together, which will induce noise and hurt the final performance. Instance-based method LEAP achieved lower performance than temporal method RETAIN, which proves the importance of temporal information in EHR data. Compared to other baselines, G-BERT has more capacity and achieves comparable results, better than GAMENet, which adding the extra information of DDI knowledge and procedure codes. The result proves the effectiveness of Transformer-based model on the medication recommendation task. Unfortunately, G-BERT ignores the temporal information, which means the model cannot learn the disease progression information, so G-BERT indicates suboptimal results.

Table 3
Performance comparison of different medication recommendation methods.

Methods	Jaccard	F1	PR-AUC
LR	0.4075 ± 0.0005	0.5658 ± 0.0002	0.6716 ± 0.0006
GRAM	0.4176 ± 0.0004	0.5788 ± 0.0005	0.6638 ± 0.0003
LEAP	0.3921 ± 0.0006	0.5508 ± 0.0004	0.5855 ± 0.0004
RETAIN	0.4456 ± 0.0001	0.6064 ± 0.0006	0.6838 ± 0.0003
GAMENet-	0.4401 ± 0.0026	0.5996 ± 0.0008	0.6672 ± 0.0005
GAMENet	0.4555 ± 0.0002	0.6126 ± 0.0009	0.6854 ± 0.0007
G −BERT$_{G^-,P^-}$	0.4186 ± 0.0009	0.5796 ± 0.0006	0.6649 ± 0.0002
G −BERT$_{G^-}$	0.4299 ± 0.0005	0.5903 ± 0.0004	0.6771 ± 0.0009
G −BERT$_{P^-}$	0.4236 ± 0.0008	0.5844 ± 0.0008	0.6704 ± 0.0005
G-BERT	0.4565 ± 0.0003	0.6152 ± 0.0005	0.696 ± 0.0007
TAHDNet(w/o DDI)	0.468 ± 0.0004	0.6264 ± 0.0005	0.6974 ± 0.0011
TAHDNet	**0.4909 ± 0.0010**	**0.6478 ± 0.0007**	**0.7285 ± 0.0018**

In our method, we compare TAHDNet and its variant TAHDNet (w/o DDI). Without the knowledge of Drug-Drug Interaction (DDI), TAHDNet (w/o DDI) is also superior to other methods, which proves that the overall architecture of the model proposed in this paper is useful.

In general, TAHDNet strives to avoid these drawbacks by learning as much information as possible from the EHR data while avoiding the introduction of noise. Firstly, we take a page from Transformer-base methods to learn global dependency information of medical ontologies and employ an unsupervised pre-training procedure on larger abundant data. Using the Transformer block, we enrich the model representations without introducing a bulk of parameters. Secondly, we use a 1D-CNN block to learn local dependencies rather than the attention mechanism. With no need for aggregating all visits together, the model avoids inducing too much noise. Finally, we treat temporal information by simulating the disease progression for each patient. Most existing works (e.g. RETAIN) are based on the assumption that the information monotonously decays according to the length of time interval, ignoring the fact that some disease progression is cyclical and time decay is different for patients with different diseases. These methods generate time information that is not suitable enough for accurate predicting. As a result, we use the fused time decay function and time-aware attention mechanism to learn the temporal attention to avoid this problem. The comparison results prove the validity of our model.

5.5. Ablation study

In this section, we use three variants of TAHDNet to demonstrate the effectiveness of each block of our method. Table 4 shows the performance of the variants and the complete TAHDNet. The settings and metrics are the same as the experiments in Section 5.2.

TAHDNet$_{P^-}$ is a variant that initializes the parameters of the model without Pre-training. TAHDNet$_{C^-}$ means TAHDNet without the 1D-CNN block, where we generate the new representation without local dependency information. TAHDNet$_{T^-}$ means TAHDNet without the Time-aware block, where we use data without time intervals as input and generate new representation without time-aware attention.

As shown in Table 4, TAHDNet$_{P^-}$ performs worse than the other variants, proving the significance of learning global dependency information by employing an unsupervised pre-training procedure on larger abundant data. Comparing the complete TAHDNet, the performance of TAHDNet$_{C^-}$ and TAHDNet$_{T^-}$ also drops, which validates that learning local dependency information and temporal information are also essential for the medication recommendation task. TAHDNet$_{T^-}$ has the least degradation in performance. The reason is the average visit number of the multiple visits data is 2.36, which means we lack enough long temporal sequences to learn time information to enhance the final representation of patients.

Table 4
Average Performance for Variants of TAHDNet.

Methods	Jaccard	F1	PR-AUC
TAHDNet	**0.4909**	**0.6478**	**0.7285**
TAHDNet$_{P^-}$	0.4196	0.5805	0.6529
TAHDNet$_{C^-}$	0.4464	0.607	0.6892
TAHDNet$_{T^-}$	0.4598	0.6196	0.6932

5.6. Parameter influence

In order to study the parameter sensitivity of the model and explore the prediction performance of the model with different embedding dimensions, the parameter search experiments are conducted in this paper. Fig. 3, Fig. 4 and Fig. 5 respectively describe the influence of parameter changes on Jaccard, F1 and PR-AUC. The results are shown in Fig. 4. It can be seen from the results in the Fig. 3, Fig. 4 and Fig. 5 that when the dimension is 300, the performance of the model reaches the maximum. Therefore, this paper sets the embedding dimension to 300.

5.7. Case study

We selected two patients from the test dataset for model recommendation performance validation: there are diagnoses and medications change among visits. In Table 5, the patient 1 has 3 diagnoses for the 1_{st} visit and two extra diagnoses Joint pain, Hyperlipidemia for the 2_{nd} visit. The ground truth medications prescribed by doctors and recommended medications by TAHDNet are listed in the table. Overall, TAHDNet recommends 13 correct, 12 correct medications for two visit respectively. In addition, we found that in the 1_{st} visit and the 2_{nd} visit, THADNet (w/o DDI) recommended medications N02B (Analgesics and Antipyretics) and C10A (Lipid-modifying Agents) can lead to harmful side effect such as Myoma. while, THADNet considered the impact of Drug Drug Interactions (DDI), so C10A was removed at the end.

In Table 6, the patient 2 has 2 diagnoses for the 1_{st} visit and 3 diagnoses Surg compl-heart, Prim cardiomyopathy and Atrial fibrillation for the 2_{nd} visit. The ground truth medications prescribed by doctors and recommended medications by TAHDNet are listed in the table. Overall, TAHDNet recommends 4 correct, 5 correct medications for two visit respectively. In addition, In the 1_{st} visit, THADNet (w/o DDI) recommended medications A01A (Stomatological Preparations) and N03A (Antiepileptic Drugs) could cause allergic bronchitis. In the 2_{nd} visit, THADNet (w/o DDI) recommended medications N02B (Analgesics and Antipyretics) and C10A (Lipid-modifying Agents) can lead to harmful side effect such as Myoma. While, THADNet considered the impact of Drug Drug Interactions (DDI), so C10A and N03A was removed respectively at the end.

On the other hand, we also assessed the interpretability of the model by selecting the visit records of patient 2 and calculating the contribution of the variable (disease diagnosis in this case) to the medication recommendation according to equation. (14). Table 7 depicts the contribution of disease diagnosis to medication recommendation in two records of patient 2: Hypothyroidism 0.006, Cardiac dysrhythmias 0.203, Surg compl-heart 0.177, Atrial fibrillation 0.225, and Prim Cardiomyopathy 0.389. From Table 7 we can conclude that the diagnosis of disease at the 2_{nd} visit and the diagnosis of arrhythmia at the 1_{st} visit

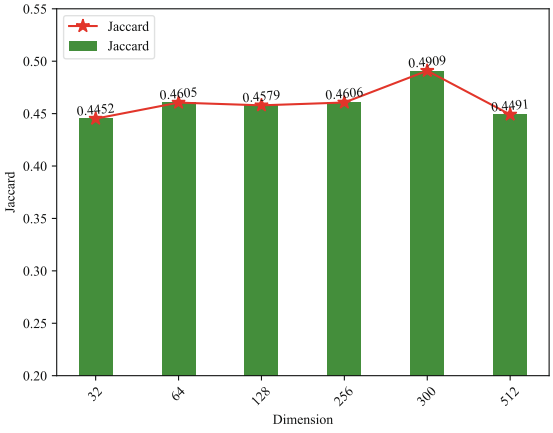


Fig. 3. The influence of parameter changes on Jaccard.

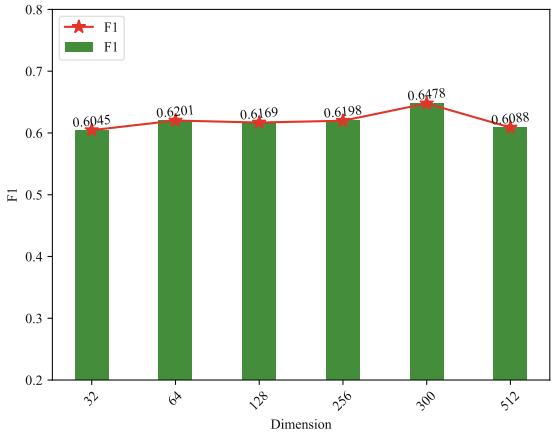


Fig. 4. The influence of parameter changes on F1.

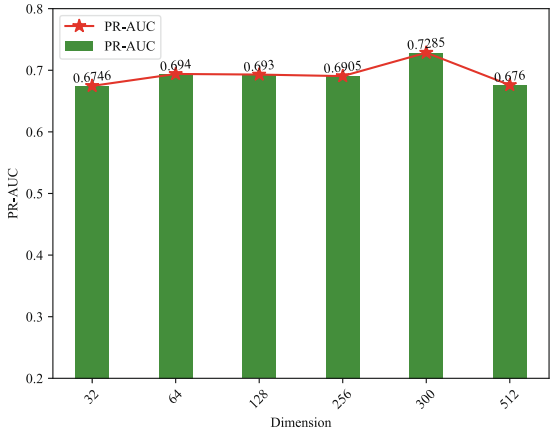


Fig. 5. The influence of parameter changes on PR-AUC.

Table 5
Example Recommended Medications for Patient 1 with Two Visits. Here "unseen" indicates the medications predicted but are not in ground truth, while "missed" refers to the medications that are in the ground truth but are not predicted.

Diagnosis	Methods	Recommended Medications
1st Visit: Secundum atrial sept def; Hypotension; Other diseases of circulatory system	Ground Truth	N02B, A01A, A02B, A06A, A12A, B05C, A12C, C01C, A07A, A10A, M01A, N01A, N07A, C02D, N02A
	TAHDNet (w/o DDI)	11 correct (N02B, A01A, A02B, A06A, A12A, B05C, A12C, C01C, A10A, N01A, N07A, N02A) + 1 unseen (C10A) + 4missed
	TAHDNet	13 correct (N02B, A01A, A02B, A06A, A12A, B05C, A12C, C01C, A07A, A10A, N01A, N07A, N02A) + 2 missed
2nd Visit: Hypotension; Joint pain; Hyperlipidemia	Ground Truth	N02B, A01A, A02B, A06A, B05C, A12A, A12C, C01C, A07A, A10A, M01A, N01A, C07A, N07A
	TAHDNet (w/o DDI)	11 correct (N02B, A01A, A02B, A06A, A12A, C01C, A07A, A10A, N01A, C07A, N07A) + 1 unseen (C10A) +3 missed
	TAHDNet	12 correct (N02B, A01A, A02B, A06A, A12A, C01C, A07A, A10A, M01A, N01A, C07A, N07A) + 2 missed

Table 6
Example Recommended Medications for Patient 2 with Two Visits. Here "unseen" indicates the medications predicted but are not in ground truth, while "missed" refers to the medications that are in the ground truth but are not predicted.

Diagnosis	Methods	Recommended Medications
1st Visit: Hypothyroidism; Cardiac dysrhythmias	Ground Truth	H03A, D11A, N05C, A04A, C10A, A01A
	TAHDNet (w/o DDI)	4 correct (H03A, D11A, N05C, A01A) + 1 unseen (N03A) + 2 missed
	TAHDNet	5 correct (H03A, D11A, N05C, C10A, A01A)+ 1 missed
2nd Visit: Surg compl-heart; Prim cardiomyopathy; Atrial fibrillation	Ground Truth	C07A, N02B, B01A, B05C, C10A, D11A, C03C, H03A, C01D, C02D, A03B
	TAHDNet (w/o DDI)	8 correct (C07A, N02B, B01A, B05C, C10A, C03C, C01D, A03B) + 1 unseen (C10A)+ 3 missed
	TAHDNet	9 correct (C07A, N02B, B01A, B05C, D11A, C03C, C01D, C02D, A03B) + 2 missed

Table 7
The Contribution of variables (diagnosis) in patient 2's visit records to medication recommendation.

Visit record	Diagnoses	Contribution
1st visit	Hypothyroidism	0.006
	Cardiac dysrhythmias	0.203
2nd Visit	Surg compl-heart	0.177
	Atrial fibrillation	0.225
	Prim cardiomyopathy	0.389
Medication Recommendation	C07A, N02B, B01A, B05C, D11A, C03C, C01D, C02D, A03B	

contributed positively to the medication recommendation. However, the diagnosis of Hypothyroidism in the 1_{st} visit was almost non-contributory to medication recommendation.

6. Conclusion

In this paper, we propose a dynamic time-aware hierarchical dependence network, TAHDNet, for the medication recommendation task. By learning local and global dependency information, the hierarchical dependencies from EHR data of a patient is obtained. In addition, We also propose a fused time decay function to stimulate the disease progression and a time-aware attention mechanism to learn attention weights for each visit for a more precise representation. Experimental results on real-world EHR data show that TAHDNet outperforms the other methods on the medication recommendation task.

In the actual medication recommendation process, we don't consider many drugs-some are part of the acute phase and some are ongoing, which is something we need to look into further.

CRediT authorship contribution statement

Yaqi Su: Methodology, Software, Validation, Writing - original draft. **Yuliang Shi:** Data curation, Supervision, Writing - review & editing. **Wu Lee:** Data curation, Software, Validation. **Lin Cheng:** Software, Validation, Supervision, Writing - review & editing. **Hongmei Guo:** Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported by the National Key Research and Development Plan of China (No.2018YFC0114709).

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